

Program: B. Tech. in Computer Science & Engineering

Semester Course Code Name of the course L T P Credits

VI CS6TXXX React.js Development 2 0 0 2

Prerequisites for the course

Knowledge of HTML, CSS, and JavaScript (ES6+)

Basics of Web Development and DOM manipulation

Prior Reading Material/useful links

<https://react.dev/learn>

<https://developer.mozilla.org/en-US/docs/Web/JavaScript>

<https://roadmap.sh/react>

Course Outcomes

Sr. No Course Outcome number CO statement

1 CO1 Student will be able to understand the fundamentals of React and JSX.

2 CO2 Student will be able to design and develop functional UI components using hooks and props.

3 CO3 Student will be able to manage application state using useState, useEffect, and Context API.

4 CO4 Student will be able to integrate APIs and handle routing in React applications.

5 CO5 Student will be able to build, deploy, and optimize a full-scale single-page React application.

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Syllabus

Unit Course Contents Hours

Unit I React Basics and JSX: Introduction to React, Advantages of React, Virtual DOM, Creating React App, Understanding JSX, Components (Class vs Functional), Props, Rendering elements, React Developer Tools. 06

Unit II State and Event Handling: useState, Event handling, Conditional rendering, Lists and Keys, Forms in React, Controlled vs Uncontrolled components, useEffect, Lifecycle in Functional components. 06

Unit III Routing and Component Communication: React Router (v6), Route Params, Nested Routes, Navigation, Passing Data via Props, Context API, useContext, Custom Hooks (Intro). 06

Unit IV API Integration and Real-World Features: Fetching data using fetch/axios, Asynchronous programming with Promises, API error handling, Loading states, Pagination, Search, Filters. LocalStorage, Custom Hooks (in detail). 06

Unit V Project Development and Deployment: Building a complete project (e.g., Blog, E-commerce UI), Component structure, Styling in React (CSS Modules / Styled Components), Environment Variables, Deployment on Netlify/Vercel, Code splitting, Performance optimization. 06

Text Books

Alex Banks & Eve Porcello, Learning React, 3rd Edition, O'Reilly, 2023

Kirupa Chinnathambi, Learning React, 2nd Edition, Addison-Wesley, 2018

Reference Books

Robin Wieruch, The Road to React, 2022 Edition

Dave Ceddia, Pure React

<https://react.dev> (Official Docs)

Useful Links

<https://react.dev>

<https://github.com/facebook/react>

<https://www.freecodecamp.org/news/learn-react-by-building-a-simple-app/>

<https://docs.netlify.com/>